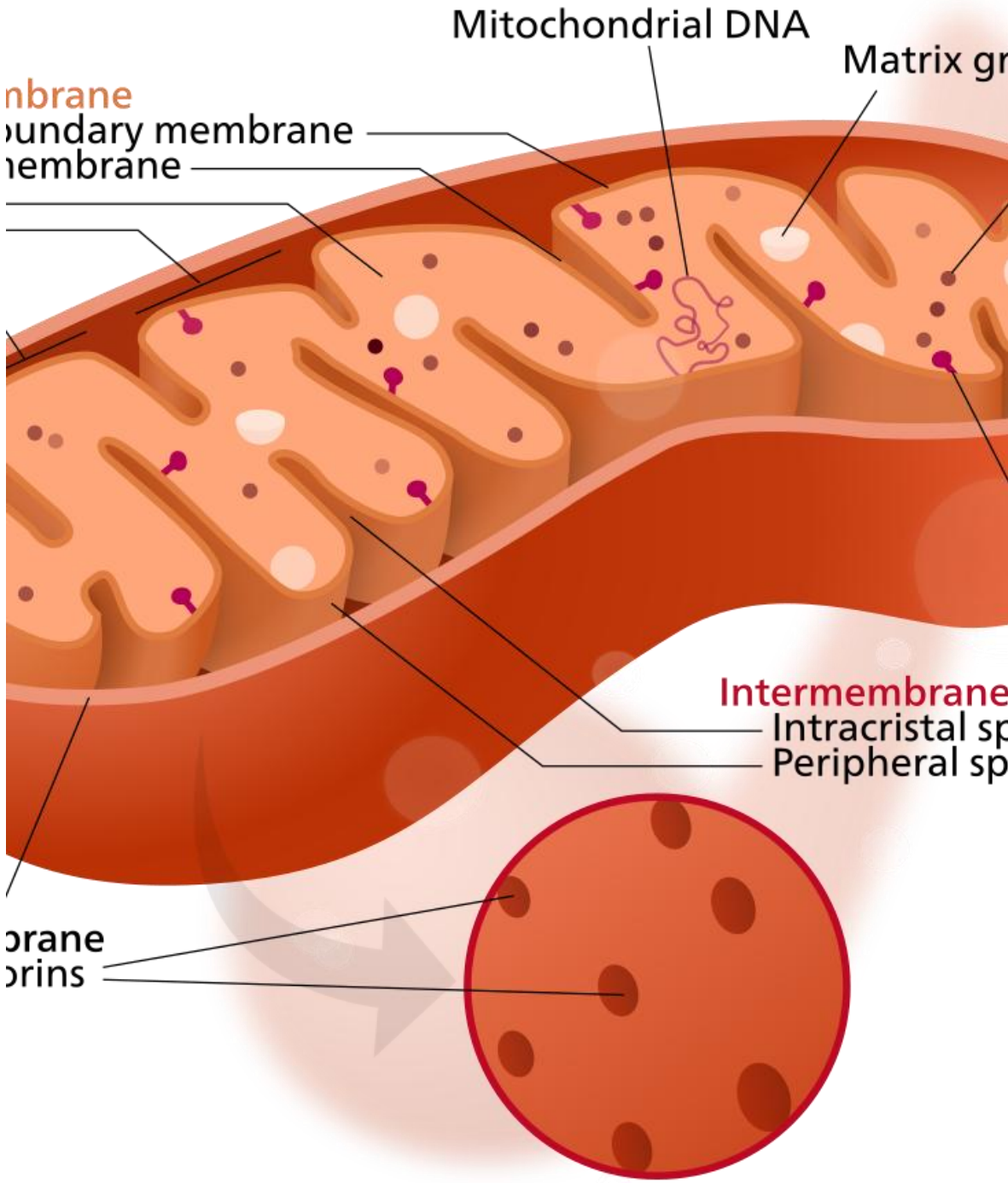
DP Biology – B2.2 (HL extension)

Organelles built for the job

- How does the structure of each organelle suit its function?
- Why is it useful for cells to have separate compartments?



HL statements B2.2.4–9. Textbook pp. 248–255.
Image: a mitochondrion. Kelvinsong, modified by Sowlos, CC BY-SA 3.0.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	B2.2.4–6	Explain how mitochondria, chloroplasts and the nuclear envelope are adapted to their functions
2	B2.2.7–9	Explain how free ribosomes, rough ER, the Golgi apparatus and vesicles produce and move proteins

The big picture: compartments with jobs

Energy

Mitochondria and chloroplasts: membranes for gradients

Control

Nucleus: envelope and pores

Export

Rough ER, Golgi, vesicles

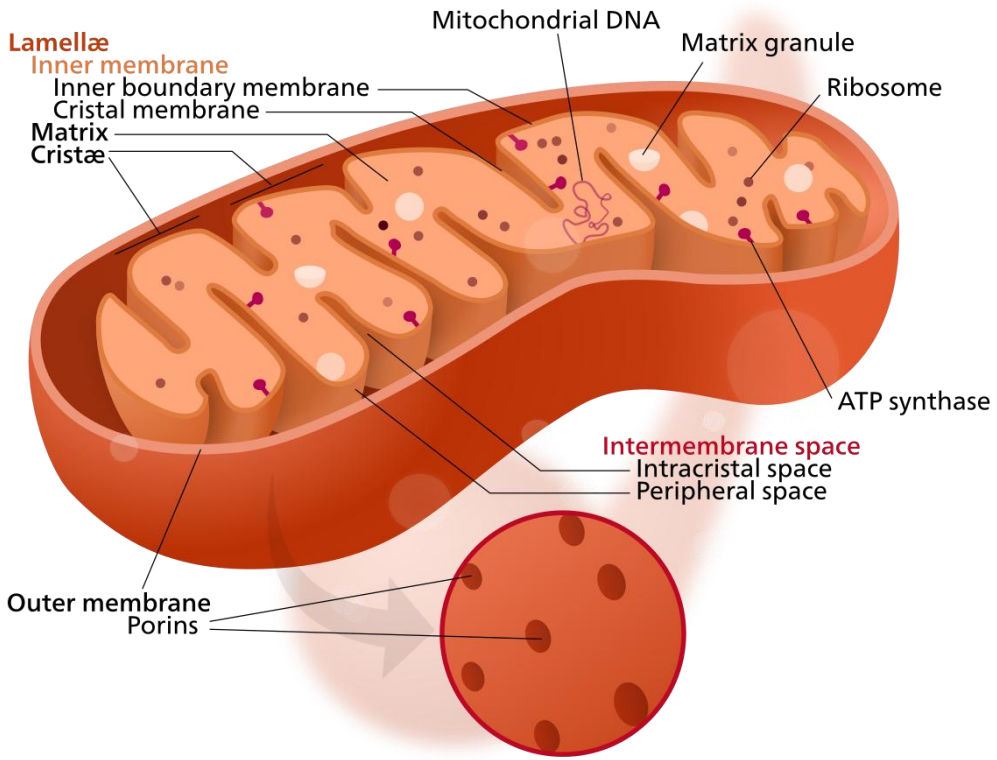
Each organelle is a **compartment** with its own conditions, and its **structure** suits its **function**.

Big question

Why do heart muscle cells and pancreas cells look so different inside?

The mitochondrion: built to make ATP

B2.2.4



Mitochondrion. Kelvinsong, modified by Sowlos, CC BY-SA 3.0.

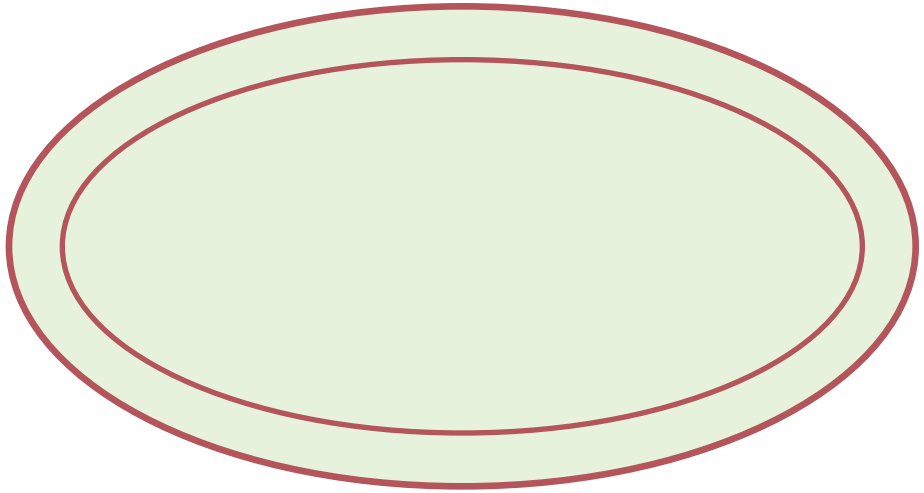
Structure	Adaptation
Double membrane	Separates respiration from cytoplasm
Small intermembrane space	Steep proton gradient fast
Cristae	Large area for ETC and ATP synthase
Matrix	Krebs cycle enzymes concentrated

Key idea

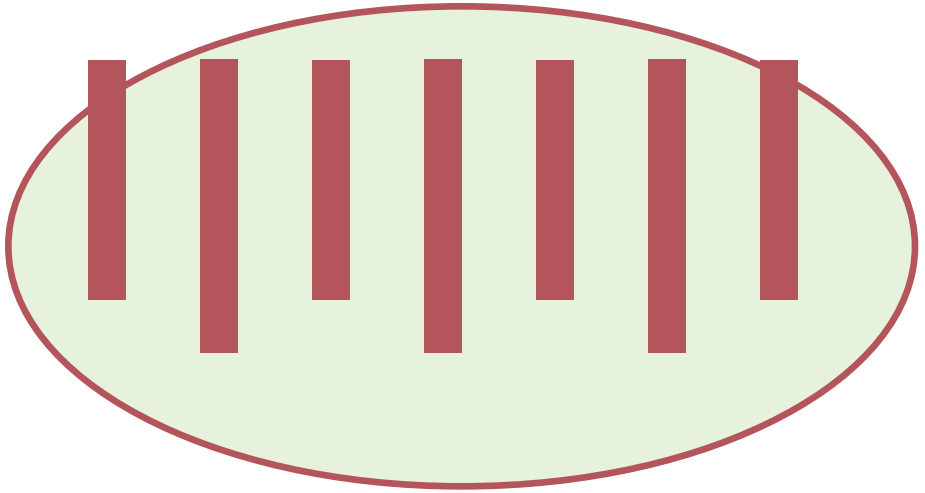
Compartments let different conditions sit side by side.

Why fold the inner membrane?

B2.2.4



no cristae: small inner membrane area



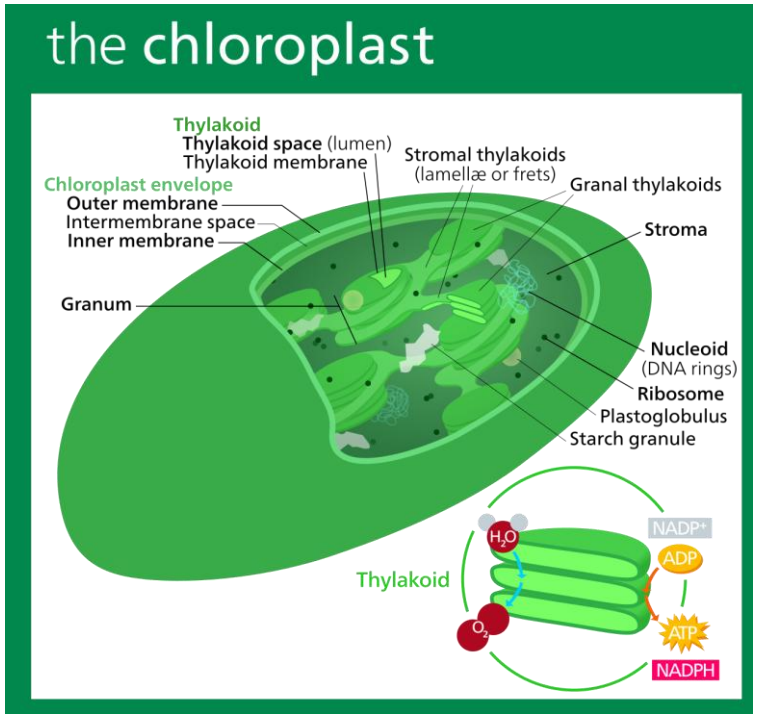
cristae: many times the area

So

More inner membrane = more electron transport chains and ATP synthase = more ATP per mitochondrion.

The chloroplast: built for photosynthesis

B2.2.5



Chloroplast. Kelvinsong, CC BY 3.0.

Structure	Adaptation
Thylakoids in grana	Large area for photosystems
Small thylakoid space	Steep proton gradient fast
Stroma	Calvin cycle enzymes (rubisco)
Double membrane	Separates photosynthesis

Compare

Find three similarities with the mitochondrion.

Two powerhouses compared

B2.2.4–5

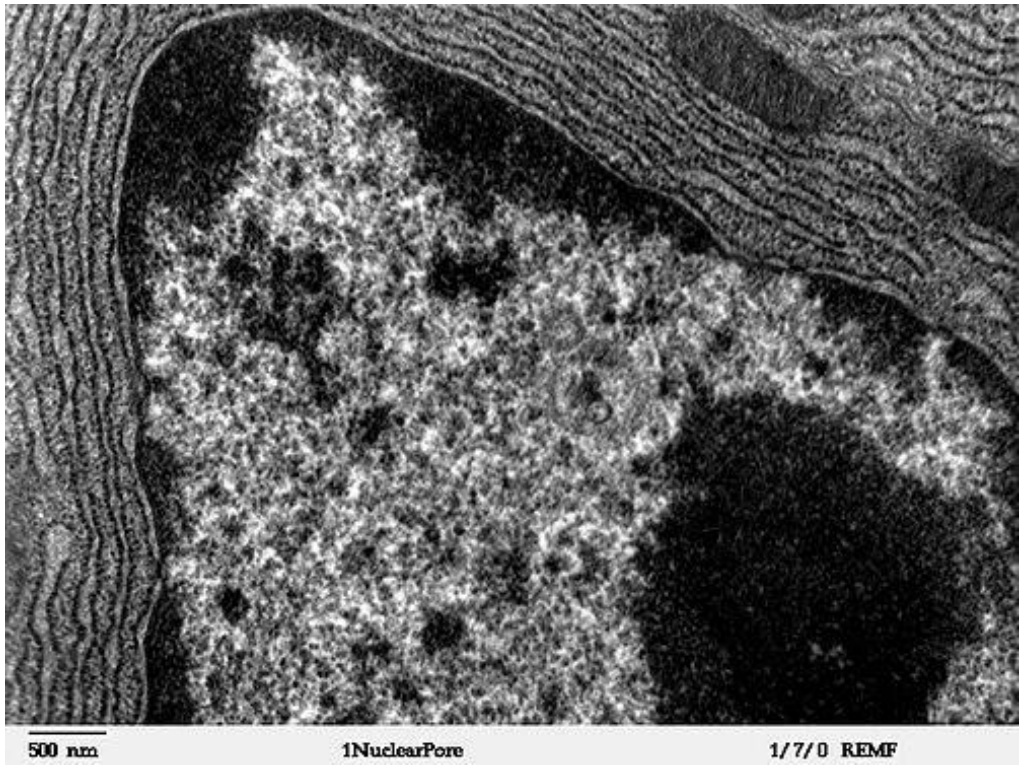
	Mitochondrion	Chloroplast
Where protons build up		
Membrane with ATP synthase		
Fluid with a cycle		
Area increased by		

Activity

Complete in pairs. Explain why a small proton space speeds ATP production.

Why the nucleus has a double membrane

B2.2.6



Nucleus edge with rough ER (TEM). Louisa Howard, public domain.

Pores control traffic: mRNA out; proteins and nucleotides in.

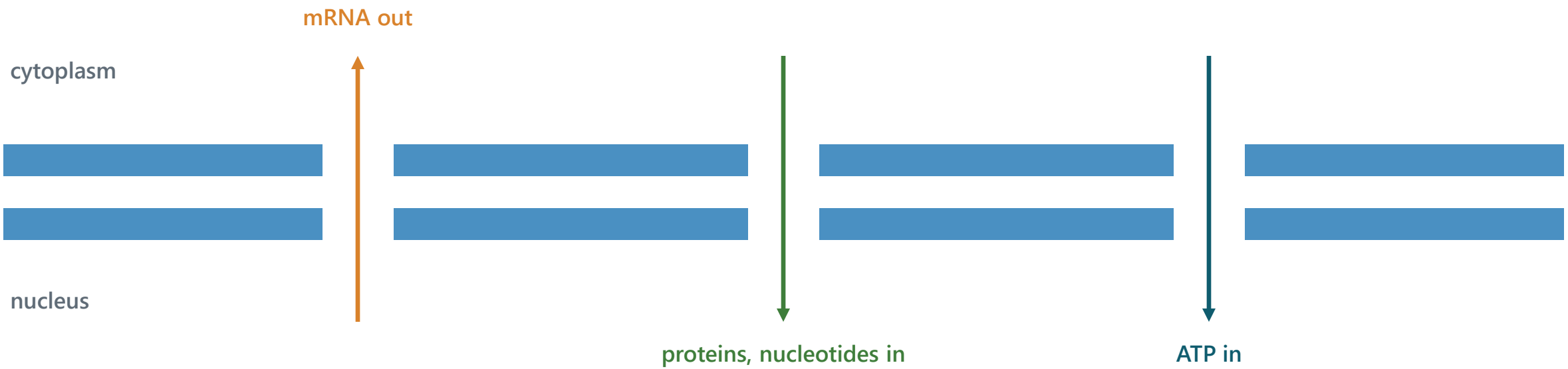
The outer membrane continues as rough ER. DNA is kept apart, separating transcription from translation.

Cell division

The envelope breaks into vesicles so spindles reach the chromosomes, then re-forms.

Traffic through nuclear pores

B2.2.6



Key idea

DNA never leaves; the pores select what crosses.

Working with data: mitochondria in different cells

B2.2.4

Percentage of cell volume that is mitochondria (rounded typical figures from micrographs).

Cell type	Mitochondria / %
Heart muscle	35
Liver cell	20
Skeletal muscle	5
Red blood cell	0

1. Why does heart muscle have the most?
2. Why do red cells have none?
3. Predict: sperm midpiece vs head.

When organelles fail

B2.2.4–9

Organelle	What can go wrong
Mitochondria	Faulty mtDNA cuts ATP; muscles and nerves hit hardest
Lysosomes	A missing enzyme: substances build up (Tay-Sachs)
Rough ER	Misfolded CFTR destroyed (cystic fibrosis)
Nuclear envelope	Faulty lamins: rapid ageing (progeria)

Think

Why are muscle and nerve cells hit hardest by mitochondrial disease?

Check-in: class 1

B2.2.4–6

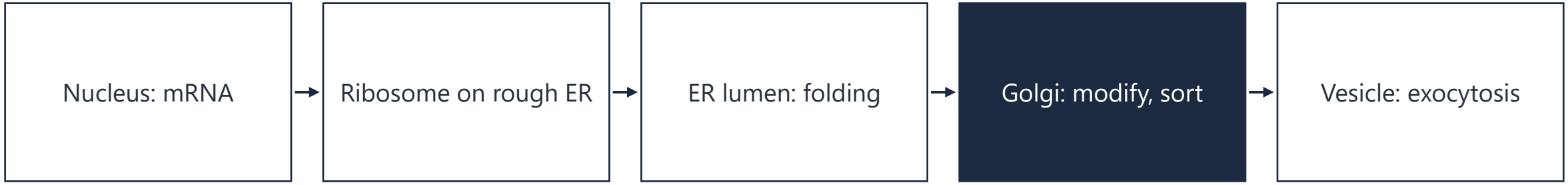
1. Explain how cristae suit the mitochondrion.
2. Explain why a small intermembrane space helps.
3. Explain how thylakoids suit the chloroplast.
4. State what passes through nuclear pores.
5. Explain why the envelope breaks down in mitosis.

How to use this

Answer without your notes, then check against slides 3–11.

The road out of the cell

B2.2.7–9



Proteins for **export** follow one route. Proteins for the **cytoplasm** are made on free ribosomes and stay put.

Example

Pancreas cells make digestive enzymes on rough ER, process them in the Golgi and secrete them.

Free ribosomes and rough ER

B2.2.7

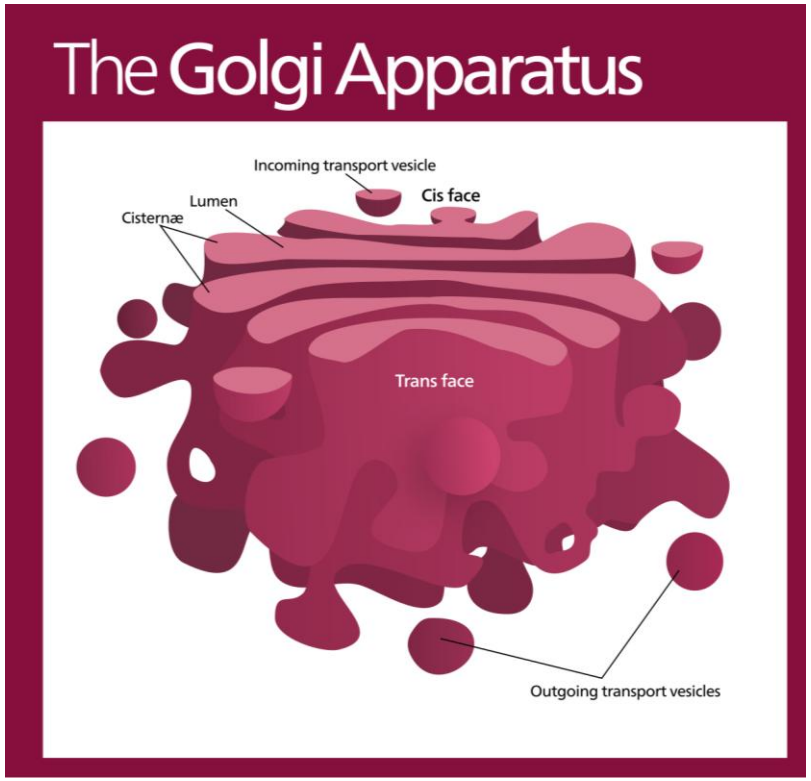
	Free ribosomes	On rough ER
Where	Cytoplasm	ER membrane
Proteins for	Cytoplasm (e.g. glycolysis enzymes)	Secretion, lysosomes, membranes
Protein goes	Into the cytoplasm	ER lumen, then Golgi
Type	80S	80S

Polysomes

Many ribosomes translate one mRNA at once: many copies fast.

The Golgi apparatus

B2.2.8



Golgi apparatus. Kelvinsong, CC BY 3.0.

A stack of **cisternae**. Vesicles from the ER fuse with one face.

Enzymes **modify** proteins (adding sugars or phosphate) as labels, then vesicles bud off the far face to their destinations.

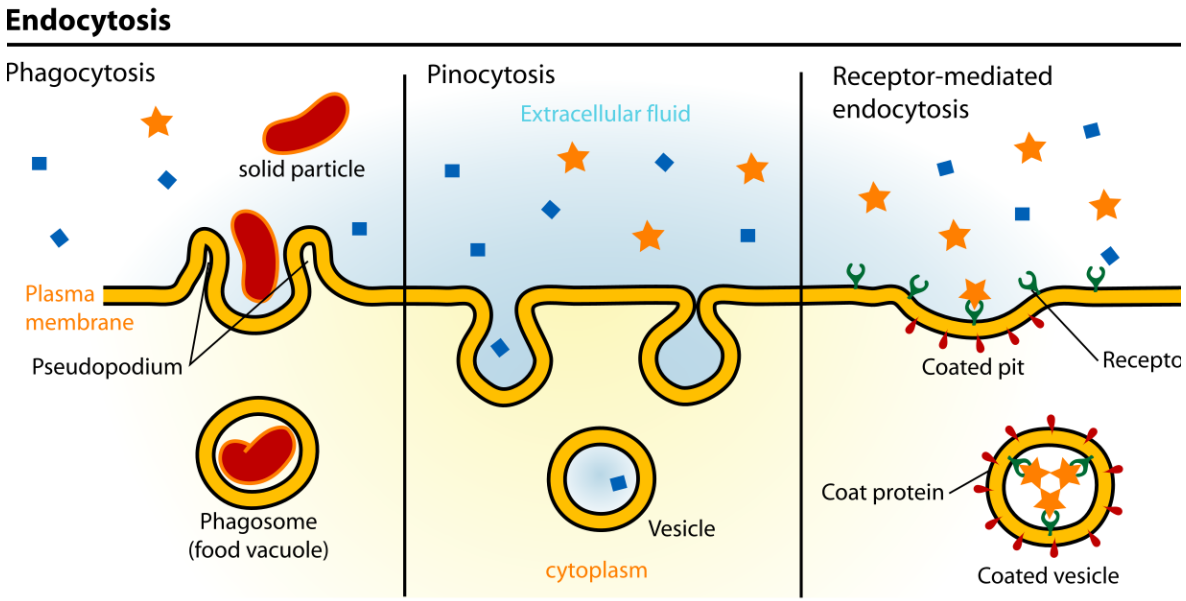
Common mistake

The Golgi does not make proteins; it modifies and sorts them.

Vesicles and clathrin

B2.2.9

Substance binds receptors → **Clathrin cage forms inside** → Membrane pulled into a pit → Pit pinches off → Coat removed; vesicle moves



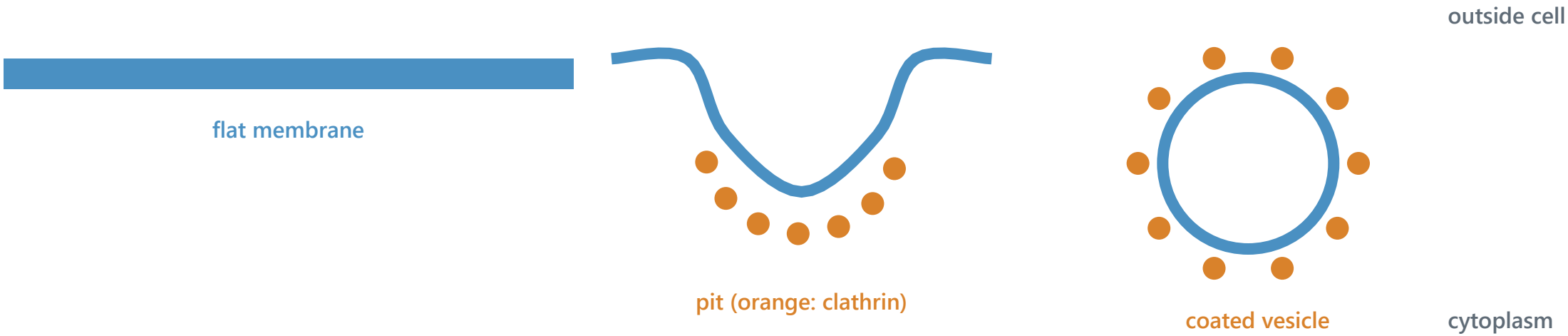
Phagocytosis, pinocytosis, receptor-mediated endocytosis. M. Ruiz Villarreal, public domain.

Uses

Cells take in LDL cholesterol, iron and hormones this way; viruses exploit it too.

A coated pit forming

B2.2.9



Key idea

Clathrin links into a cage that bends the membrane into a vesicle.

Follow a protein out of the cell

B2.2.7–9

1. Draw the route of an antibody from gene to outside a plasma cell.
2. **Predict** where a protein ends up if it loses its ER signal.
3. **Suggest** why secretory cells have lots of rough ER and Golgi.

Route
Nucleus → rough ER ribosome → ER lumen → vesicle → Golgi → vesicle → exocytosis.

Practice: protein traffic

B2.2.7–9

Distinguish

Free vs bound ribosome products.

Outline

Two roles of the Golgi.

Describe

How clathrin forms a vesicle.

Use

rough ER, cisternae, glycosylation, clathrin

Check-in: class 2

B2.2.7–9

1. Distinguish proteins made on free and bound ribosomes.
2. Outline the role of the Golgi apparatus.
3. Describe the role of clathrin.
4. State two uses of receptor-mediated endocytosis.
5. Explain why polysomes are useful.

How to use this

Answer without your notes, then check against slides 13–19.

Common mistakes to avoid

Not this	But this
The Golgi makes proteins.	It modifies and sorts them.
Nuclear pores let DNA out.	mRNA and proteins pass; DNA stays.
Cristae increase volume.	They increase inner membrane area.

Summary: organelles built for the job

B2.2.4–9

	Key idea
B2.2.4	Mitochondria: cristae, small space, matrix
B2.2.5	Chloroplasts: thylakoids, stroma
B2.2.6	Nuclear envelope: pores; breaks in division
B2.2.7	Free vs bound ribosomes
B2.2.8	Golgi modifies and sorts
B2.2.9	Clathrin-coated vesicles

Check that you can...

1. **Explain** adaptations of mitochondria and chloroplasts
2. **Explain** the nuclear envelope and pores
3. **Distinguish** free and bound ribosomes
4. **Outline** the Golgi and vesicles
5. **Describe** clathrin vesicle formation

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. State why a small intermembrane space helps ATP production.
2. Distinguish between proteins made by free ribosomes and by ribosomes on the rough ER.
3. Outline the role of clathrin.

Image credits

Mitochondrion structure.svg – Kelvinsong; modified by Sowlos – CC BY-SA 3.0

Chloroplast II.svg – Kelvinsong – CC BY 3.0

Mammalian pancreatic tissue, nuclear pores and rough endoplasmic reticulum.jpg – Louisa Howard – Public domain

Golgi apparatus (standalone version)-en.svg – Kelvinsong – CC BY 3.0

Endocytosis types.svg – Mariana Ruiz Villarreal LadyofHats – Public domain

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